

Current through $3\text{ }\Omega$ $= 18 / 4 = 4.5\text{ A}$

23 B The p.d. across LM is less than e.m.f. of the standard cell. Thus, if resistance of LM increased, its p.d. will increase to a value beyond e.m.f. of the standard cell, resulting in a balance point.

24 A As the top pan balance's reading increases when there is a current in direction XY in the wire